

Data Structures: Assignment-1

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Q1: An array is a bitonic array if all integers from index 0 to index i are sorted in ascending order, and all subsequent integers from index i+1 to n-1 are sorted in descending order. Given a bitonic array of n distinct integers, write a C/C++ program to find the maximum integer in the array in $O(\log(n))$ time.

```
1  #include <iostream>
2  using namespace std;
3
4  int findMax(int A[], int n) {
5      int st=0,end=n-1,mid;
6      while(st<=end) {
7          mid=st+(end-st)/2;
8          if(A[mid-1]>A[mid]) {
9              end=mid-1;
10         }
11         else if(A[mid+1]>A[mid]) {
12             st=mid+1;
13         }
14         else {
15             return A[mid];
16         }
17     }
18     return -1;
19 }
20
21 int main() {
22     int n;
23     cout << "Enter the number of elements: ";
24     cin >> n;
25     int A[n];
26     cout << "Enter the elements: ";
27     for(int i=0; i<n; i++) {
28         cin >> A[i];
29     }
30     cout << "The maximum number in the array is: " << findMax(A,n);
31 }
32
```

Output:

```
PS E:\Assignment-1> cd "e:\Assignment-1\" ; if ($?) { g++ Q1.cpp -o Q1 } ; if ($?) { .\Q1 }
Enter the number of elements: 6
Enter the elements: 1 2 4 8 7 6
The maximum number in the array is: 8
```

Q2: Let $A[n]$ be an array of n distinct integers. If $i < j$ and $A[i] > A[j]$, then the pair (i, j) is called an inversion of A . Write a C/C++ program that determines the number of inversions in any permutation on n elements.

```
1  #include <iostream>
2  using namespace std;
3
4  int countInversion(int A[],int n) {
5      int i,j,count=0;
6
7      for(i=0; i<n; i++) {
8          for(j=i+1; j<n; j++) {
9              if(A[i]>A[j]) {
10                 count++;
11             }
12         }
13     }
14
15     return count;
16 }
17
18 int main() {
19     int n;
20     cout << "Enter the number of elements: ";
21     cin >> n;
22     int A[n];
23     cout << "Enter the elements: ";
24
25     for(int i=0; i<n; i++) {
26         cin >> A[i];
27     }
28
29     cout << "The number of inversions are: " << countInversion(A,n);
30 }
31
```

Output:

```
PS E:\Assignment-1> cd "e:\Assignment-1\" ; if ($?) { g++ Q2.cpp -o Q2 } ; if ($?) { .\Q2 }
Enter the number of elements: 4
Enter the elements: 4 1 3 2
The number of inversions are: 4
```

Q3: Write a C program to manage the details of students using an array of structures.

The program should:

1. Accept the number of students (n) from the user.
2. For each student, input the following details:
 - Roll number (integer)
 - Name (string)
 - Marks (floating-point value)
3. Store the details of all students in an array of structures.
4. Display the details of all students in a formatted way.

```
1  #include <stdio.h>
2
3  struct student {
4      int roll;
5      char name[100];
6      float marks;
7  };
8
9  int main() {
10     int n;
11     printf("Enter the number of students: ");
12     scanf("%d",&n);
13     struct student arr[n];
14
15     for(int i=0; i<n; i++) {
16         printf("Enter details of Student %d\n", i+1);
17         printf("Enter roll number: ");
18         scanf("%d",&arr[i].roll);
19         printf("Enter name: ");
20         scanf("%s",&arr[i].name);
21         printf("Enter marks: ");
22         scanf("%f",&arr[i].marks);
23     }
24
25     for(int i=0; i<n; i++) {
26         printf("\nDetails of Student %d\n", i+1);
27         printf("Roll no: %d\n",arr[i].roll);
28         printf("Name: %s\n",arr[i].name);
29         printf("Marks: %f\n",arr[i].marks);
30     }
31 }
32
```

Output:

```
PS E:\Assignment-1> cd "e:\Assignment-1\" ; if ($?) { gcc Q3.c -o Q3 } ; if ($?) { .\Q3 }
Enter the number of students: 2
Enter details of Student 1
Enter roll number: 101
Enter name: Akash
Enter marks: 396
Enter details of Student 2
Enter roll number: 102
Enter name: Parth
Enter marks: 480

Details of Student 1
Roll no: 101
Name: Akash
Marks: 396.000000

Details of Student 2
Roll no: 102
Name: Parth
Marks: 480.000000
```